

Graph Neural Networks for Relational Learning: A Survey

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Abstract

This paper presents a survey of recent advances in the field. We review key contributions, discuss methodological trends, and identify open challenges. Our analysis covers both foundational work and recent developments, highlighting directions for future research.

1. Introduction

Graph-structured data is ubiquitous in real-world applications. Early work on graph neural networks introduced the notion of node-level state propagation [5]. Graph convolutional networks [1] extended spectral methods to efficient spatial convolutions, enabling scalable semi-supervised node classification. Graph attention networks [2] introduced learnable per-edge weights, improving expressivity. For large-scale inductive settings, neighbourhood sampling approaches [3] allow generalization to unseen nodes at inference time.

2. Related Work

The expressiveness of GNNs has been formally analyzed in relation to the Weisfeiler-Leman graph isomorphism test [4]. Spectral graph convolutions using adaptive filters have been extended to dynamic networks [6], while hyperbolic embedding spaces have been applied to knowledge graph completion [7]. Differentiable pooling methods [8] enable hierarchical graph representations. Contrastive objectives have been applied to molecular graphs [9], and symmetry-equivariant architectures have been proposed for physical systems [10].

References

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